Testing Multi-modal LLM Judges of Educational Content: Effectiveness, Robustness, and Fair Representation

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***Abstract*—Educational assessment providers are increasingly adopting multi-modal large language models (VLMs) both to generate image-based test items and to evaluate them before human review (MLLM-as-a-Judge). This paradigm poses significant societal and operational risks: a judge that misses defects may expose students to invalid items or stereotype threat, whereas one that over-flags items wastes reviewers' time. Testing such judges is difficult because no ground-truth oracle exists for their verdicts, particularly for culturally dependent traits where geometric transformations cannot inject meaningful defects.**

To evaluate MLLM judges empirically, we define six quality properties for image-based multiple-choice questions, grounded in educational theory and validated with an industrial partner. We test two production VLMs across 1,940 items from ScienceQA using five metamorphic operators (three severity-graded parametric operators and two image-substitution operators), an unmodified control, and counterfactual stereotype-tagged portraits for Fair Representation, totaling 68,868 judgments.

On unmodified items, baseline flag rates are below 1.5% for technical and presentation properties, 0% for Fair Representation, and 14.6–23.1% for Functional Relevance. Recall on injected defects ranges from 0.51 (rotation) to 0.90 (blur). While recall increases with severity for clutter and blur, it drops from 78–95% at 15° rotation to 26–37% at 90°, driven by canvas-border artefacts at oblique angles. Furthermore, evaluating all six properties in a single prompt achieves comparable recall to single-property prompting (0.741 vs. 0.757; $p = 0.006$) while reducing evaluation costs 2.3-fold. Finally, stereotype-tagged substitutes are flagged in 23.1% of evaluations, but 94% of these flags occur on a single concept pairing (weapons), demonstrating how explicit object detection can confound representational fairness testing.

***Index Terms*—LLM-as-a-judge, metamorphic testing, test oracle, fairness testing, vision-language models, educational assessment**

I. Introduction

Standardised educational tests inform consequential decisions about students' academic trajectories. The quality and fairness of individual test items directly affect both the validity of score interpretations and educational equity [1, Chs. 4 and 12]. Exposing students to construct-irrelevant visual defects or culturally biased depictions can induce stereotype threat and systematically disadvantage specific demographic groups [2]. However, producing high-quality, unbiased items manually requires considerable time and effort from assessment providers and domain experts [3].

To scale item creation, assessment providers are integrating large language models (LLMs) and vision-language models (VLMs) into their authoring pipelines both to generate candidate items and to review them [4]. Because generative models are prone to hallucinations and societal biases [5], providers rely on the LLM-as-a-Judge paradigm [6] as an automated quality gate before human review. Within this pipeline, the judge acts as a software component with a well-defined contract: it takes an item as input and outputs a pass/fail verdict per quality property. Yet LLM judges are themselves fallible [7]. A missed defect allows invalid or unfair content to reach students, whereas false alarms consume reviewers' time and erode trust in the pipeline. Consequently, multi-modal judges must be systematically tested before deployment.

This need emerged directly from our collaboration with an industrial partner, a large educational assessment provider. While established taxonomies define quality criteria for text-only items [8]–[10], no standard rubric covers items where an image carries core information. We therefore defined six quality properties for image-based multiple-choice questions (MCQs)—*Functional Relevance*, *Visual Clarity*, *Technical Quality*, *Standard Presentation*, *Text–Image Coherence*, and *Fair Representation*—grounded in the multimedia effect in testing [11], cognitive load theory [12], and visual literacy [13]. Our partner's domain experts reviewed and adopted these six properties without modification for their production pipeline.

Evaluating whether a VLM judge enforces these properties reliably is an instance of the test oracle problem [14]. First, we evaluate the judges on curated, unmodified items to measure baseline agreement and potential false-alarm rates. Second, to measure defect detection, we apply metamorphic testing [15] using property-specific operators—including severity-graded visual transformations—to inject defects with known expected verdicts. Testing *Fair Representation*, however, requires a distinct approach: geometric transformations cannot inject social stereotypes, and prompting generative models to synthesize biased imagery for testing is both unreliable and ethically contentious. Instead, we substitute item images with portraits from an existing counterfactual social-bias dataset (CMSC) [16] and compare verdicts against the unmodified baseline.

We evaluate two production VLMs acting as zero-shot judges across 1,940 image-based items from ScienceQA [17], executing 68,868 judgments across four research questions:

- **RQ1 (Effectiveness).** How consistently do multi-modal LLM judges recognise unmodified educational items as satisfying the six quality properties?

- **RQ2 (Robustness).** Can the LLM judges detect artificially injected quality defects in image-based educational items, and how does detection vary with defect severity?
- **RQ3 (Cost-effectiveness).** How does combined-property prompting compare with single-property prompting in defect detection and cost per complete item evaluation?
- **RQ4 (Fair Representation).** How do the judges' Fair Representation verdicts differ between items with stereotype-tagged substitute images and unmodified items?

Our results show that on unmodified items, judges flag Technical Quality and Standard Presentation in at most 1.4% of evaluations and Fair Representation in 0%, but flag Functional Relevance in 14.6–23.1% (RQ1). For injected defects, targeted recall spans from 0.51–0.63 (rotation) to 0.85–0.90 (blur), increasing monotonically with severity only for clutter, while 15° rotations are detected far more often (78–95%) than 90° rotations (26–37%) due to canvas border artefacts (RQ2). Combined-property prompting reduces evaluation cost 2.3-fold with only a 1.6-percentage-point drop in overall recall compared to single-property prompting (RQ3). Finally, while Fair Representation flags 23.1% of stereotype-tagged substitutes versus 0% of unmodified items, 94% of flags fall on a single pairing depicting weapons (RQ4), highlighting critical confounding factors when designing fairness tests for multi-modal judges.

In summary, this paper makes the following contributions:

1) Six quality properties for image-based MCQs, grounded in educational literature and validated with an industrial partner.
2) A metamorphic testing framework using five visual and semantic operators, an unmodified baseline control, and a collateral flag analysis.
3) Empirical evidence from 68,868 judgments across two production VLMs on 1,940 educational items, including a paired cost-effectiveness comparison of prompting strategies.
4) A non-generative evaluation design for Fair Representation that reuses counterfactual datasets, exposing how object-level confounders impact fairness testing.
5) An open-source, resumable evaluation harness (Section IX).

II. Background and Related Work

A. LLMs for Item Generation and Quality Assurance

Multiple-choice questions (MCQs) are widely used in standardised testing for their scoring efficiency and diagnostic versatility [18]. An MCQ comprises a *question stem* and a set of options: one correct *key* and plausible incorrect *distractors*.

Because manual item authoring is costly [3], automated MCQ generation has been studied since the 1990s [19], progressing from rule-based NLP [20], [21] to modern LLMs used for stem and distractor synthesis in industrial pipelines [4], [22], [23]. However, generative models frequently produce hallucinations [5], including cross-modal inconsistencies where a question references details absent from its accompanying image [24].

To filter defective outputs, pipelines employ the LLM-as-a-Judge paradigm [6], which can correlate well with human assessment when guided by explicit rubrics [25]. While evaluation rubrics exist for text-only items [8], [9], [26], they do not address visual quality or cross-modal alignment. We bridge this gap in Section III.

Crucially, defining a rubric does not guarantee reliable enforcement. LLM judges exhibit positional and verbosity biases [6], and multi-modal judges struggle with absolute scoring consistency [27]. Furthermore, when evaluating multiple criteria simultaneously, LLM judges can suffer from criteria confusion, allowing defects in one dimension to bleed into verdicts on others [28].

Recent studies have begun perturbing inputs to probe multi-modal judges: MM-JudgeBias measures compositional biases across queries and images [29], Park et al. apply visual perturbations to mitigate perceptual bias [30], and Wang et al. craft adversarial perturbations to inflate safety and quality scores [31]. Rather than studying general-purpose preference bias or adversarial evasion, we inject targeted domain defects into educational items to test whether judges accurately flag violated pedagogical properties while isolating undamaged ones.

The closest prior work evaluated 17 LLMs on 14 text-only item properties using text mutations, identifying image-based item evaluation as open future work [10]. We address that multi-modal extension here.

B. Testing Components Without an Oracle

When expected outputs are unknown for arbitrary inputs, metamorphic testing verifies relations across transformed inputs [15], [32], while fault injection supplies inputs with known defects [33]. Both rely on a *partial* oracle that constrains specific outputs while leaving others open [14]. In multi-property judging, a mutation targeting property p defines the expected verdict (fail) for p alone; it does not guarantee that property $q \neq p$ remains unaffected. Treating untargeted flags as false positives risks misclassifying valid collateral detections.

While visual transformations such as blur, rotation, and object insertion are established in testing autonomous driving [34], corruption robustness [35], and image captioning [36], [37], and metamorphic testing has been applied to socially critical software [38], we adapt these operators to test multi-property quality gates where collateral cross-property effects must be explicitly separated from baseline flag rates.

C. Fairness Testing and Social Bias in VLMs

Software fairness is a core software engineering concern [39], with automated fairness testing typically probing whether decision systems discriminate across protected attributes [40]–[43]. Testing a judge *of* representational fairness inverts this setup: positive test cases must themselves depict social stereotypes. Because text-to-image models amplify societal stereotypes [44], frameworks like ImageBiTe automate bias testing against ethical requirements [45], [46],

while counterfactual datasets like SocialCounterfactuals [47] and CMSC [16] vary demographic attributes across social concepts. We repurpose CMSC to evaluate whether VLM judges flag stereotyped depictions in educational items without synthesizing new harmful imagery.

III. Quality Properties for Items with Images

We synthesized six quality properties from research on the multimedia effect in testing [11], cognitive load theory [12], and visual literacy [13]. We documented these properties in a formal specification and reviewed them with domain experts at our industrial partner to verify alignment with their human review guidelines. The experts adopted all six properties without modification:

- **Functional Relevance (Rel.):** The image provides information necessary to answer the question rather than serving as pure decoration [11], [48].
- **Visual Clarity (Clar.):** The image presents required elements cleanly, avoiding extraneous visual clutter that increases cognitive load [49], [50].
- **Technical Quality (Tech.):** Resolution, contrast, and legibility suffice to perceive all relevant details [1].
- **Standard Presentation (Pres.):** The image and its internal elements appear in conventional, canonical orientations [50].
- **Text–Image Coherence (Coh.):** The question text and accompanying image convey mutually consistent information [12], [51].
- **Fair Representation (Fair):** The image avoids cultural assumptions and stereotypes that could advantage, disadvantage, or alienate groups of students [1], [2], [52].

While the first five properties capture perceptual and semantic validity, *Fair Representation* directly addresses societal equity. To operationalize Fair Representation in our prompts, we incorporate the 19 generic validation conditions from ImageBiTe [46] (e.g., exaggerated stereotypes, associating crime or danger with a demographic group, colonial or exoticising depictions), instructing the judge that matching any single condition fails the property.

IV. Study Design

Figure 1 illustrates our experimental pipeline. Each evaluation unit is a tuple (*item, operator, severity, model, strategy*). The operator transforms or substitutes the item's image, the VLM judge evaluates the resulting item, and the output verdicts are compared against the partial oracle.

A. Research Questions

RQ1 evaluates judges on unmodified ScienceQA items under the baseline assumption that curated benchmark items satisfy our six properties, quantifying baseline disagreement and reviewer workload. **RQ2** measures whether judges detect defects injected into specific properties and how recall scales with defect severity, alongside a supporting analysis of collateral flags on untargeted properties. **RQ3** compares single-property prompting against combined-property prompting to test whether evaluating six criteria simultaneously causes cross-property interference or offers a viable cost-reduction strategy. **RQ4** evaluates whether Fair Representation verdicts distinguish stereotype-tagged portraits from unmodified items and examines the methodological boundary conditions of substitution-based fairness testing.

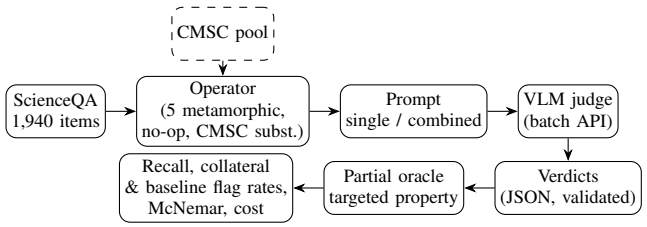


Fig. 1. Study pipeline. Each unit (item, operator, severity, model, strategy) is fingerprinted, cached, and submitted through a batch API.

Listing 1. Prompt template (Jinja2). The image is attached as a separate input.

```
Consider the IMAGE attached. Given the CHARACTERISTICS
     below and the
QUESTION, evaluate whether the IMAGE complies with each of
     the
following LIST_OF_CONDITIONS.
CHARACTERISTICS: Grade: {{grade}}, Subject: {{subject}},
     Topic: {{topic}}
QUESTION: {{question}}
LIST_OF_CONDITIONS:
{% for prop in properties %}- {{prop.name}}:
     {{prop.definition}}
{% endfor %}
For each condition, respond in JSON format as a list of
     objects, each
with these exact keys:
- "property": the property name
- "passed": true or false
- "reasoning": a brief explanation if passed is false,
     otherwise an
  empty string
Respond with ONLY the JSON, no other text.
```

B. Judges and Prompting Strategies

Judges are queried zero-shot with a single user message containing the rendered prompt template (Listing 1) and the base64-encoded image. The prompt supplies the item's grade, subject, topic, question stem, and property definitions. Answer options and explanations are excluded so the judge evaluates the direct alignment between the question stem and image.

In *combined-property prompting*, the prompt lists all six properties simultaneously. In *single-property prompting*, it lists only the single property targeted by the active operator (requiring six separate API calls for a full item evaluation). Responses are parsed and validated against a strict JSON schema requiring boolean verdicts and non-empty reasoning on failures; malformed responses are discarded and logged.

C. Metamorphic Operators

Table I summarizes our operators. All follow the same metamorphic relation: assuming an unmodified item satisfies property p, applying the operator targeting p yields a defective item that a sound judge must fail on p.

The three *parametric* operators degrade visual properties across three severity levels (Figure 2): *Blur* applies a Gaussian

TABLE I

METAMORPHIC OPERATORS. SEVERITY VALUES ARE THE PARAMETERS USED IN THE CODE; SUBSTITUTION OPERATORS HAVE NO SEVERITY PARAMETER.

Target property	Operator	Subtle	Moderate	Obvious
Technical Quality	Gaussian blur, radius (px)	2	6	15
Standard Presentation	Rotation, angle	15°	90°	180°
Visual Clarity	Clutter: circles / size[a]	2 / 5%	5 / 10%	10 / 20%
Functional Relevance	Random substitution		none	
Text–Image Coherence	Topic substitution[b]		none	
Fair Representation	CMSC substitution		none	
(all)	No-op control		none	

[a] Diameter as a fraction of the shorter image side. [b] Same subject, different topic.

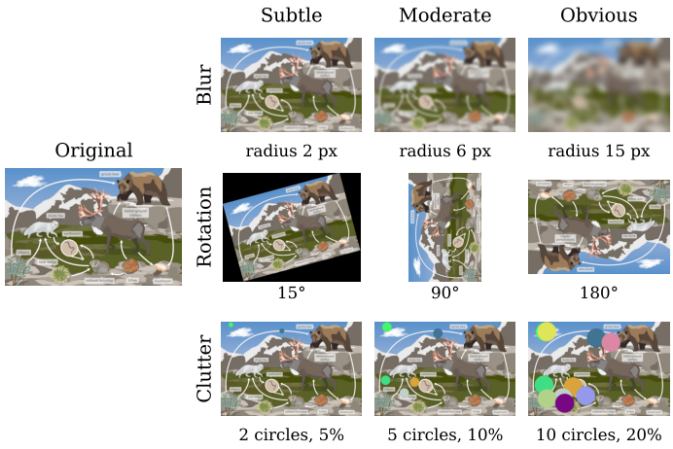


Fig. 2. A ScienceQA item image and the output of the three parametric operators at each severity, generated with the study's operator code. Rotation expands the canvas and fills the corners (visible at 15°).

filter; *Rotation* rotates the image and expands the canvas bounds to prevent cropping; and *Clutter* overlays randomly positioned, solid-colored circles whose count and diameter scale with severity.

The two *substitution* operators replace the item's image with another image from ScienceQA using a deterministic seed derived from the global seed (42) and the item ID. *Random substitution* draws from the entire dataset, whereas *Topic substitution* draws from items within the same subject but a different topic, preserving domain plausibility while breaking question-image coherence. Because substitution operators have no severity parameter, we utilize the three severity slots in the pipeline to execute three independent runs on byte-identical inputs at default temperature. We exclude these repeats from severity-scaling analyses and instead leverage them in Section VII to quantify intra-judge verdict consistency across identical inputs.

D. Fair Representation: Substitution and Control

To test Fair Representation without synthesizing new harmful content, we substitute item images with portraits from CMSC [16], a counterfactual dataset depicting social concepts across varied demographic attributes. Because our judge evaluates single items in isolation rather than dataset-wide distributions, we selected four concept–demographic pairings representing documented societal stereotypes visible in individual portraits: *bad student* (young Black men, 25 images), *weapon* (young Black men, 35 images), *belligerent* (Middle Eastern men, 48 images), and *unpleasant* (Black women, 48 images), forming a pool of 156 images. Each item is assigned a random image from this pool using its item-specific seed.

Because a naive judge could achieve 100% recall on substitutes simply by flagging every image, we pair this experiment with a *no-op control* evaluating all 1,940 unmodified items under combined prompting. This control establishes both the specificity of Fair Representation on standard educational visuals and the baseline flag rates for RQ1 and collateral analysis.

TABLE II

EXPERIMENTAL PHASES AND NUMBER OF PARSED JUDGMENTS. EACH JUDGMENT IS ONE REQUEST; A COMBINED REQUEST RETURNS SIX VERDICTS.

Ph.	Operators	Items	Strategy	Models	Judgments
1	5 metamorphic × 3 sev.	194	single	luna	2,910
1	5 metamorphic × 3 sev.	194	combined	luna	(in Ph. 2)
2	5 metamorphic × 3 sev.	1,940	combined	luna, terra	58,199
3	CMSC substitution	1,940	combined	luna, terra	3,880
4	No-op control	1,940	combined	luna, terra	3,879
Total					68,868

All combined prompts use the 19-item Fair Representation checklist (Section III). Phase 2: 29,100 (luna) and 29,099 (terra) parsed judgments. Phase 4: one response failed validation.

E. Subjects and Experimental Phases

We use all 1,940 image-containing items from the ScienceQA test split [17], spanning natural, social, and language sciences across elementary, middle, and high school grades.

We evaluate two production OpenAI VLMs configured in our industrial partner's evaluation harness: `gpt-5.6-luna` (luna) and `gpt-5.6-terra` (terra),[1] queried via the Batch API with default generation parameters. luna and terra differ by a factor of ten in token pricing ($0.20/$1.20 vs. $2.00/$12.00 per million input/output tokens, including the 50% Batch API discount). A planned second provider's batch endpoint was unavailable during the experimental window, restricting our comparison to two tiers within one model family (Section VII).

Table II details the four experimental phases totaling 68,868 parsed judgments. Phase 1 compares single-property and combined-property prompting on a 10% subject-stratified sample (194 items) using luna. Phase 2 evaluates all 1,940 items across the five metamorphic operators using combined prompting on both luna and terra (reusing cached Phase 1 combined evaluations). Phases 3 and 4 execute the CMSC substitution and unmodified no-op control across all 1,940 items for both models.

[1] Exact model identifiers configured in the partner evaluation harness, representing a cost-optimised tier (luna) and a flagship high-compute tier (terra).

F. Oracle and Metrics

For RQ1, we report the *baseline flag rate* on unmodified items; its complement represents agreement with the assumed-valid benchmark.

For RQ2, each operator defines the expected verdict (fail) solely for its targeted property p. We report targeted *recall* (fraction of mutated units flagged on p). Because a mutation can legitimately degrade untargeted properties $q \neq p$ (e.g., substituting an image breaks both Relevance and Coherence, and heavy blur degrades Visual Clarity), we do not classify untargeted flags as false positives.[2] Instead, we report the *collateral flag rate* on $q \neq p$ and compare it against the unmodified baseline flag rate from Phase 4.

For RQ3, we compare paired binary detection outcomes on the 2,910 Phase 1 units using McNemar's test [53] with continuity correction, alongside 95% bootstrap confidence intervals (1,000 resamples) [54] and exact API token costs.

G. Execution Infrastructure

Each unit is uniquely fingerprinted by hashing its item ID, model, operator, severity, seed, strategy, and rendered prompt. A persistent cache and batch ledger ensure idempotency and crash recovery, while a run manifest locks code revisions and configurations. The entire harness is unit-tested and open-sourced.

V. Results

A. RQ1: Effectiveness on Unmodified Items

Table III presents the baseline flag rates on the 1,940 unmodified ScienceQA items (Phase 4). Both judges exhibit high agreement with the baseline on Technical Quality (1.0–1.4% flagged), Standard Presentation (0.2%), and Fair Representation (0.0%). Visual Clarity and Text–Image Coherence show moderate flag rates between 4.2% and 9.9%.

Functional Relevance exhibits the highest baseline disagreement, flagged in 23.1% of items by luna and 14.6% by terra. Two factors contribute to this rate: first, some curated benchmark items genuinely contain decorative images that are not strictly required to answer the question; second, because our prompt omits the MCQ answer options (Section IV), items where the visual link depends on the options rather than the stem alone may appear functionally irrelevant to the judge. In an industrial pipeline, a 15–23% baseline flag rate on Relevance indicates that prompts must either include full item options or treat Relevance flags as soft warnings to avoid overwhelming human reviewers.

B. RQ2: Robustness to Injected Defects

Table IV reports recall on the targeted property for combined prompting on all 1,940 items, per operator and, for the parametric operators, per severity.

Detection appears to be strongly dependent on the property considered. Blur is detected in 85–90% of evaluations, random substitution in 89%, topic substitution in 85–89%, clutter in 59–64%, and rotation in 51–63%. Severity affects the three parametric operators differently. For clutter, recall rises steeply and monotonically, from 0.26 and 0.36 with two small circles to 0.88 and 0.90 with ten large ones. For blur, recall jumps from 0.71 and 0.60 at a 2 px radius to near-ceiling at 6 px and does not increase further. For rotation, recall is not monotone: the 15° rotation is detected in 78% and 95% of evaluations, the 90° rotation in 26% and 37%, and the 180° rotation in 50% and 56%. The substitution operators have no severity parameter, and their recall is flat across the three labels (0.89–0.90 for random and 0.84–0.89 for topic substitution), as expected for identical inputs. terra has higher recall than luna for clutter, rotation, and topic substitution, at every severity for the first two. luna is higher for blur at every severity, and the two models are equal for random substitution (0.893 vs. 0.892).

We observe that the rotation result contradicts the operator's intended severity order. The 15° rotation is the only one that leaves filled corners on an expanded canvas (Figure 2), whereas 90° and 180° rotations produce a complete, rectangular image. This indicates that the judges react to this visible framing artefact rather than to the orientation itself, and that sideways or upside-down diagrams are often accepted when the image boundary appears normal. For blur, the 6 px and 15 px levels are both detected almost always, so the operator

TABLE III
RQ1: Flag rates (%) on unmodified items (combined prompting, Phase 4). Lower rates indicate greater agreement with the assumed-valid baseline.

Property	luna	terra
Functional Relevance	23.1	14.6
Visual Clarity	4.2	5.3
Technical Quality	1.4	1.0
Standard Presentation	0.2	0.2
Text–Image Coherence	5.1	9.9
Fair Representation	0.0	0.0

TABLE IV
RQ2: Recall on the targeted property (combined prompting, Phase 2). Top: severities pooled (5,820 evaluations per cell; 5,819 for rotation on terra). Bottom: parametric operators by severity (1,940 per cell; 1,939 for 90° on terra).

Operator	Severity	luna	terra
Blur (Tech.)	pooled	0.895	0.847
Clutter (Clar.)	pooled	0.585	0.643
Rotation (Pres.)	pooled	0.512	0.625
Random subst. (Rel.)	–	0.893	0.892
Topic subst. (Coh.)	–	0.846	0.885
Blur (Tech.)	2 px	0.707	0.597
	6 px	0.994	0.973
	15 px	0.986	0.970
Clutter (Clar.)	2 circles, 5%	0.261	0.356
	5 circles, 10%	0.618	0.671
	10 circles, 20%	0.878	0.903
Rotation (Pres.)	15°	0.777	0.949
	90°	0.260	0.370
	180°	0.499	0.555

[2]Computing naive precision by assuming all $q \neq p$ must pass yields 0.34–0.51 under combined prompting and 1.00 by construction under single-property prompting; neither reflects true judge precision.

distinguishes only two effective levels for these judges. Only clutter behaves as a graded operator across its whole range.

It is important to mention that severity levels are only rank changes within an operator, not across operators: a 15° rotation and a 2 px blur are not directly comparable. Furthermore, some science visuals (e.g., an object photographed from above or symmetrical shapes) have no strict canonical orientation, meaning a subset of rotated images may function as equivalent mutants.

Supporting analysis: flags on untargeted properties: Table V reports, for each operator and model, the fraction of evaluations in which each untargeted property was flagged, over all three severities (5,820 evaluations per cell, 5,819 for rotation on terra; 1,940 for the CMSC substitution). The baseline row gives the flag rate of each property on unmodified items from the control.

The two substitution operators also trigger the other semantic property: random substitution triggers Coherence in 86.9% (luna) and 89.9% (terra) of evaluations, and topic substitution triggers Relevance in 87.4% and 87.5%. Blur triggers Clarity in 68.7% and 73.8%, well above the unmodified baseline of 4.2% and 5.3% (RQ1). These patterns are plausible: substitution can damage both Relevance and Coherence, while blur can damage both Technical Quality and Clarity.

The implication for RQ2 is limited but important: flags on untargeted properties are not automatically false positives. Other effects, such as increased Coherence flags after blur, reflect either illegible visual labels breaking text-image alignment or the criteria confusion reported for text evaluators [28]. Without human labels, we cannot determine which additional flags are correct or distinguish recognition of multiple defects from a generic reaction to an abnormal image.

C. RQ3: Cost-effectiveness of Prompting Strategies

Table VI compares the strategies on the 2,910 units of Phase 1 (194 items, five operators, three severity labels), all judged by luna under both strategies.

Regarding defect detection, single-property prompting detected the targeted defect in 163 units in which combined prompting did not, and combined prompting in 116 units in which single-property prompting did not (McNemar's $\chi^2 = 7.58$, $p = 0.006$). Overall recall was 0.741 for combined and 0.757 for single-property prompting, a difference of 1.6 points. The strategies disagreed on 279 units (9.6%), so the choice of strategy changes the verdict for about one defective item in ten, even though the aggregate difference is small. The direction of the difference depends on the operator: combined prompting had higher recall for blur (+7.4 points) and lower recall for clutter (-7.9), topic substitution (-5.0), rotation (-2.3), and random substitution (-0.3). By severity, it was higher for subtle defects (+1.1) and lower for moderate (-3.1) and obvious ones (-2.9). Per operator, the difference favours combined prompting for blur (49 vs. 6 discordant units, $p < 10^{-7}$) and single-property prompting for clutter (55 vs. 9, $p < 10^{-7}$) and topic substitution (43 vs. 14, $p = 2\times10^{-4}$); for rotation (43 vs. 30, $p = 0.16$) and random substitution (16 vs. 14, $p = 0.86$) it is not significant. With a Bonferroni correction for five tests ($\alpha = 0.01$), the three significant differences remain significant. The small aggregate difference therefore combines a gain on one operator with losses on others.

In terms of cost, the 2,910 single-property requests required \$0.73 and the 2,910 combined requests \$1.92. A combined request thus cost 2.6× a single-property request rather than 6×, because the image, which dominates the input tokens, is sent once per request in either strategy. Obtaining verdicts for all six properties with single-property prompting requires six requests, i.e., \$4.36 for the same units, or 2.3× the cost of combined prompting. This assumes that single-property requests for the untested properties cost as much as those for the tested ones. Across the whole study, luna cost \$22.27 and terra \$195.06 (8.8× more) at the configured list prices, for recall that differs by at most 11.3 points (Table IV).

Consequently, asking about all six properties at once reduced overall detection by only 1.6 points. This difference is statistically detectable but small, and it does not show the large interference that motivated single-property prompting. Because the direction of the difference depends on the operator, we recommend neither strategy on the grounds of detection alone; the 2.3-fold cost reduction is the stronger argument for combined prompting, with the loss concentrated in clutter and topic substitution. While combined prompting may benefit from the comparative context of the other definitions or be more willing to fail properties overall, the supporting analysis of collateral flags cannot separate these mechanisms on this 194-item sample.

D. RQ4: Fair Representation

Table VII reports the flag rate of Fair Representation under the CMSC substitution and the control. luna failed Fair Representation in 450 of 1,940 evaluations of substituted items (23.2%) and terra in 445 of 1,940 (22.9%); neither model failed it for any of the 3,879 unmodified items. The unmodified-item result is consistent with Phase 2, where Fair Representation was flagged in at most two of about 29,100 judgments per model on items transformed by the other operators (Table V). In addition, the table splits the substitutes by CMSC pairing. Since the stored results do not record which pairing each substitute belongs to, we recovered this information by replaying the seeded draw of each substitute from the pool of 156 images. To ensure that the replay is correct, we compared it with the judges' justifications: for every flagged verdict of luna, the two agree (see below).

The result establishes that, on ordinary ScienceQA images, the judges did not raise Fair Representation flags. For a practitioner, this bounds one cost of deploying the property: on content of this kind, it will not flood reviewers with false alarms. Under CMSC substitution, the judges also failed on average about three of the other five properties (65.0% of those verdicts for luna, 60.0% for terra), mostly Relevance, Clarity, and Coherence (Table V). This is expected, since a portrait is irrelevant to, and incoherent with, a science question.

TABLE V

SUPPORTING ANALYSIS: COLLATERAL FLAG RATE (%) OF EACH PROPERTY UNDER EACH OPERATOR (COMBINED PROMPTING, ALL SEVERITIES POOLED; 5,820 EVALUATIONS PER CELL, 1,940 FOR THE CMSC SUBSTITUTION OF PHASE 3). SHADED CELLS ARE THE TARGETED PROPERTY (SEE TABLE IV). THE FAIR COLUMN GIVES ABSOLUTE COUNTS BECAUSE ALL RATES ROUND TO 0.0%. BASELINE: FLAG RATE ON UNMODIFIED ITEMS (PHASE 4).

Operator (target)	Model	Tech.	Clar.	Pres.	Rel.	Coh.	Fair (count)
Blur (Tech.)	luna	–	68.7	12.5	47.0	26.2	0
	terra	–	73.8	7.7	41.8	34.0	0
Clutter (Clar.)	luna	17.2	–	0.9	27.6	17.4	0
	terra	14.0	–	0.6	27.5	31.9	0
Rotation (Pres.)	luna	6.0	22.8	–	24.0	12.0	0
	terra	3.8	25.2	–	20.5	16.7	0
Random subst. (Rel.)	luna	2.4	65.3	8.0	–	86.9	0
	terra	0.9	67.2	3.1	–	89.9	0
Topic subst. (Coh.)	luna	2.7	63.7	9.0	87.4	–	2
	terra	1.3	67.7	2.2	87.5	–	0
CMSC subst. (Fair)	luna	3.6	98.6	25.7	98.8	98.2	–
	terra	0.7	95.4	6.7	98.7	98.6	–
Baseline (no-op)	luna	1.4	4.2	0.2	23.1	5.1	0
	terra	1.0	5.3	0.2	14.6	9.9	0

Control: 1,940 (luna) and 1,939 (terra) evaluations.

TABLE VI

RQ3: RECALL ON THE TARGETED PROPERTY BY STRATEGY (PHASE 1, LUNA, 194 ITEMS; 582 UNITS PER OPERATOR, 970 PER SEVERITY). 95% BOOTSTRAP CIS OVER UNITS.

	Single-property	Combined
Overall	0.757 [0.742, 0.773]	0.741 [0.724, 0.756]
Blur (Tech.)	0.826	0.900
Clutter (Clar.)	0.675	0.596
Rotation (Pres.)	0.540	0.517
Random subst. (Rel.)	0.895	0.892
Topic subst. (Coh.)	0.847	0.797
Subtle	0.690	0.701
Moderate	0.736	0.705
Obvious	0.844	0.815
Cost of the 2,910 requests	\$0.73	\$1.92
Cost for six-property verdicts	\$4.36	\$1.92

TABLE VII

RQ4: FAIR REPRESENTATION FLAG RATE (COMBINED PROMPTING, CHECKLIST DEFINITION). "OTHER PROPERTIES": SHARE OF THE FIVE OTHER VERDICTS THAT FAILED UNDER CMSC SUBSTITUTION. ROWS BELOW THE SUBSTITUTION ROW SPLIT ITS FLAGS BY PAIRING (FLAGGED COUNT AND RATE). ONE CONTROL RESPONSE FROM TERRA FAILED VALIDATION.

Condition	luna	terra
CMSC substitution: Fair flagged	450 / 1,940	445 / 1,940
Weapon (424 items)	420 (99.1%)	422 (99.5%)
Unpleasant (608)	23 (3.8%)	13 (2.1%)
Belligerent (604)	5 (0.8%)	3 (0.5%)
Bad student (304)	2 (0.7%)	7 (2.3%)
No-op control: Fair flagged	0 / 1,940	0 / 1,939
CMSC substitution: other properties	65.0%	60.0%
Both models: substitution	895 / 3,880 (23.1%)	
Both models: control	0 / 3,879 (0.0%)	

When considering the average, however, one must be aware of the fact that it hides a sharp split between pairings. On the one hand, the weapon pairing is flagged in 99.1% (luna) and 99.5% (terra) of its evaluations, and alone accounts for 842 of the 895 flags (94%). On the other hand, the remaining three pairings are flagged in at most 3.8% of their evaluations, even though the same images fail Relevance, Clarity, and Coherence in 95–99% of the cases (Table V). The justifications provided by luna confirm this pattern: 420 of its 450 justifications (93%, based on a keyword count) mention a weapon or an armed context, and all of them refer to weapon-pairing images. In particular, 14 of the 15 justifications we sampled describe a Black man holding a rifle in a threatening setting, and explicitly state that the image associates a racial group with danger or violence. The remaining 30 flags are not related to weapons, and we did not inspect all of them; similarly, we did not examine the justifications of terra.

We cannot consider this result a miss rate. For the three pairings that are rarely flagged, two explanations remain, and our data cannot separate them: either the judges are insensitive to the stereotype, or many of these images do not actually depict it. Indeed, we only know that each image belongs to a pairing we selected as commonly stereotyped, and traits such as being a bad student or belligerent are hard to convey in a single portrait. Human labels on a sample of the substitutes would allow us to distinguish between the two. Second, for the weapon pairing, the flags do not show that the judges recognise a racial stereotype. A judge that fails any image of an armed person in a threatening setting, regardless of the person's race, would produce the same result. The checklist definition makes this scenario plausible, since one of its 19 conditions fails an image that associates negative traits, such as crime or danger, with a group.

Closing this gap is possible by leveraging CMSC's counterfactual structure, because for every substitute, the same set contains the same composition and concept with a different depicted race. A judge that recognises the stereotype should flag the selected image substantially more often than its race-swapped counterparts; a judge that reacts to the concept or

to the presence of a person should not. Incorporating this counterfactual baseline is a primary direction for our future work.

VI. Discussion

Regarding the deployment of judges as quality gates, miss rates on the targeted property ranged from 3% or less (blur of 6 px or more) to 74% (90° rotation, luna), and small clutter was missed in more than half of the evaluations. The judges are therefore better suited as a first filter before human review than as a replacement for it, and the residual miss rate must be estimated per property and defect type: a single aggregate recall hides both the spread across properties and the rotation anomaly, where the more visible artefact, not the more severe defect, drove detection. The supporting analysis of collateral flags shows that a flag on one property may reveal issues in other properties, too. For instance, for a substituted image, the judges typically fail Relevance, Coherence, and Clarity together. Baseline flag rates matter as much: Relevance was failed for 15–23% of curated items. Reviewers should therefore treat a judge's per-property verdicts as a pointer to a defective item, not as a diagnosis, and tool builders should not route items to specialised reviewers on the basis of a single failed property. RQ3 indicates that combined prompting is the more economical choice at a small loss in detection (1.6 points), although the better strategy differs per property; its collateral flagging is a cost that has to be measured against human labels before it can be called a disadvantage. The roughly nine-fold price difference between the two models bought higher recall on three of the five operators, by up to 11.3 points, equal recall on random substitution, and lower recall on blur (−4.8 points).

From the perspective of testing multi-property judges, metamorphic testing provides expected failures for the properties known to be affected by an operator, not expected passes for every other property. For example, image substitution can affect both Relevance and Coherence. Such additional effects should not be counted automatically as false positives. This caveat concerns precision computed by labelling all untargeted properties as valid; it does not invalidate property-specific precision computed from valid originals and known-defective mutants when those labels are justified. Here, we report targeted recall, baseline flag rates, and a supporting analysis of collateral flags; human labels would be needed to establish which additional flags are errors.

Finally, when testing fairness judges without generating harmful content, reusing an existing counterfactual dataset allowed us to evaluate the Fair Representation verdict without producing any new stereotyped imagery. The same dataset also offers the control that our current design lacks, namely race-swapped counterfactuals that hold everything but the protected attribute constant. Our results show why this control matters. A comparison against unmodified items can demonstrate specificity on ordinary content, which is the practically important failure mode for reviewer workload, but not sensitivity to stereotypes, which is the failure mode that matters for students. Interestingly, the overall flag rate on the substitutes (23.1%) hides a split: the judges flag the weapon pairing in 99% of the evaluations, but the other three pairings in at most 4%, although the same images fail Relevance, Clarity, and Coherence in 95–99% of the cases. Hence, the judges reliably detect an explicit depiction of an armed person; however, we cannot tell whether they are insensitive to the other stereotypes, or whether those images simply do not depict them. To distinguish between the two, both human labels and the race-swapped comparison are needed.

VII. Threats to Validity

With respect to construct validity, our metamorphic relations assume that each original item satisfies all properties. While standard for curated benchmarks, some items contain decorative images, and omitting MCQ answer options from the prompt inflates baseline Functional Relevance flags. Operators can also introduce artefacts unrelated to the targeted property, as the filled corners of the 15° rotation do. Similarly, the substitution operators can produce equivalent mutants when the substitute happens to fit the question [33], making our substitution recall a conservative lower bound. Finally, the four stereotype pairings were selected by the authors without participatory input from the affected communities; they cover specific concept–demographic pairings but do not represent the full space of representational harms in educational content.

Concerning internal validity, the harness associates batch responses with requests by position, although each response carries its request identifier. To mitigate the risk of batch-level reordering or dropped requests, we verified that the output order matches the input order across all batches, confirming zero mismatches. Because the analysis combines results from several phases, we recalculated them separately for each model and phase, and compared the prompting strategies only on Phase 1 units that luna evaluated under both strategies. Furthermore, because each unit was evaluated at the default, non-zero temperature, individual verdicts may vary across runs. By leveraging the three identical runs per item on our non-parametric substitution operators, we quantified intra-judge consistency directly: the targeted verdict was consistent across all three evaluations for 94.4% of luna items with random substitution and 89.4% with topic substitution, while for terra, the corresponding rates were 93.2% and 90.8%. Thus, identical inputs produced inconsistent verdicts for about 6–11% of items, which limits the reliability of a single verdict for an individual item while leaving aggregate rates stable.

Regarding external validity, both evaluated models come from one provider and one model family, so our findings about model differences are limited to this pair; prior work on text-only items found large differences across model families [10]. ScienceQA items are curated and contain few depictions of people, which is favourable to a low Fair Representation false-positive rate; real items that depict people may yield more flags. The CMSC images are synthetic portraits, which may be easier to judge than photographs or illustrations found in real

items. All evaluations are zero-shot with one prompt template; other prompts may change absolute rates.

In terms of conclusion validity, McNemar's test is applied to paired binary outcomes on the Phase 1 sample only; the strategy comparison therefore rests on 194 items and one model. Bootstrap intervals resample units, which are not independent within an item.

VIII. Broader Impact and Ethical Considerations

Automated judges are being placed between generative models and the students who take the resulting tests. Their errors are not symmetric in who bears them. Missed representational harms affect the students who are depicted stereotypically or who experience stereotype threat [2], and construct-irrelevant content distorts scores for some groups more than others [1]. Excessive flagging burdens reviewers and, if visible to item writers, may discourage the inclusion of people from diverse groups in items altogether. Our study measures the second risk on ordinary content and shows that the first cannot be measured with a comparison against unmodified items alone. For practitioners, the concrete consequence is that a Fair Representation judge validated only with unmodified control data should not be relied on to detect stereotypes.

We deliberately did not generate stereotyped images. The CMSC images we used were produced and released by third parties for bias research under a CC BY-NC-ND 4.0 licence; we use them unmodified, only as test inputs, and do not redistribute them: the artifact contains the selection scripts only. The four pairings we selected encode specific stereotypes about Black and Middle Eastern people; selecting them was necessary to construct positive test inputs, but our results should not be read as evidence about fairness detection for other groups or other forms of misrepresentation (e.g., disability, religion, socioeconomic status). The authors selected the pairings informally and no one from the affected communities was involved. This is a first scan with automated judges; studies with users are planned for a subsequent study, and we name the absence of community input as a limitation. No human participants were involved.

IX. Conclusion

We defined and validated six quality properties for image-based educational test items and evaluated two production VLM judges across 68,868 judgments on 1,940 ScienceQA items. On unmodified items, judges rarely flagged technical, presentation, or fairness properties, but flagged Functional Relevance in 15–23% of cases. Under metamorphic fault injection, targeted recall ranged from 0.51 (rotation) to 0.90 (blur), revealing that oblique rotation detection is driven by canvas border artefacts rather than image orientation, and that single mutations frequently cause collateral failures across semantically coupled properties. Combined-property prompting reduced evaluation costs 2.3-fold with only a 1.6-percentage-point drop in recall compared to single-property prompting. Finally, substituting stereotype-tagged portraits yielded a 0% baseline flag rate on unmodified items and a 23.1% flag rate on substitutes—94% of which were driven by weapon depictions—highlighting the need for counterfactual object controls in fairness testing. Future work will incorporate race-swapped counterfactual baselines, human-labeled collateral subsets, and cross-provider VLM evaluations.

Data Availability

The harness, operators, prompts, and analysis scripts are available at https://anonymous.4open.science/r/fair-judges-artifact-645E/. Judge responses and derived tables are included. ScienceQA is publicly available [17]; CMSC images must be obtained from the original release [16].

References

[1] American Educational Research Association, American Psychological Association, and National Council on Measurement in Education, *Standards for Educational and Psychological Testing*. Washington, DC: American Educational Research Association, 2014.

[2] C. M. Steele and J. Aronson, "Stereotype threat and the intellectual test performance of African Americans," *Journal of Personality and Social Psychology*, vol. 69, no. 5, pp. 797–811, 1995.

[3] G. Kurdi, J. Leo, B. Parsia, U. Sattler, and S. Al-Emari, "A systematic review of automatic question generation for educational purposes," *International Journal of Artificial Intelligence in Education*, vol. 30, pp. 121–204, 2020.

[4] R. Degiovanni and J. Cabot, "Towards reliable LLM-based exam generation: Lessons learned and open challenges in an industrial project," in *Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE), Industry Showcase Track*, 2025.

[5] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, and T. Liu, "A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions," *ACM Transactions on Information Systems*, vol. 43, no. 2, pp. 42:1–42:55, 2025.

[6] L. Zheng *et al.*, "Judging LLM-as-a-judge with MT-bench and chatbot arena," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.

[7] A. S. Thakur, K. Choudhary, V. S. Ramayapally, S. Vaidyanathan, and D. Hupkes, "Judging the judges: Evaluating alignment and vulnerabilities in LLMs-as-judges," in *Proceedings of the Fourth Workshop on Generation, Evaluation and Metrics (GEM) at ACL*, 2025, pp. 404–430.

[8] T. M. Haladyna, S. M. Downing, and M. C. Rodriguez, "A review of multiple-choice item-writing guidelines for classroom assessment," *Applied Measurement in Education*, vol. 15, no. 3, pp. 309–333, 2002.

[9] M. J. Gierl and H. Lai, "Evaluating the quality of medical multiple-choice items created with automated processes," *Medical Education*, vol. 47, no. 7, pp. 726–733, 2013.

[10] R. Degiovanni, S. Morales, M. Coccia, R. Clarisó, and J. Cabot, "Robust LLM-as-a-judge validators for assessing the quality of educational exams," in *Proceedings of the 41st ACM/SIGAPP Symposium on Applied Computing (SAC)*, 2026.

[11] L. Hu, G. Chen, P. Li, and J. Huang, "Multimedia effect in problem solving: A meta-analysis," *Educational Psychology Review*, vol. 33, no. 4, pp. 1717–1747, 2021.

[12] J. Sweller, J. J. G. van Merriënboer, and F. G. W. C. Paas, "Cognitive architecture and instructional design," *Educational Psychology Review*, vol. 10, no. 3, pp. 251–296, 1998.

[13] S. Stokes, "Visual literacy in teaching and learning: A literature perspective," *Electronic Journal for the Integration of Technology in Education*, vol. 1, no. 1, pp. 10–19, 2002.

[14] E. T. Barr, M. Harman, P. McMinn, M. Shahbaz, and S. Yoo, "The oracle problem in software testing: A survey," *IEEE Transactions on Software Engineering*, vol. 41, no. 5, pp. 507–525, 2015.

[15] S. Segura, G. Fraser, A. B. Sánchez, and A. Ruiz-Cortés, "A survey on metamorphic testing," *IEEE Transactions on Software Engineering*, vol. 42, no. 9, pp. 805–824, 2016.

[16] H. Cheng, Y. Guo, Q. Guo, M. Yang, T. Gan, W. Guan, and L. Nie, "Social debiasing for fair multi-modal LLMs," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 1740–1750.

[17] P. Lu, S. Mishra, T. Xia, L. Qiu, K.-W. Chang, S.-C. Zhu, O. Tafjord, P. Clark, and A. Kalyan, "Learn to explain: Multimodal reasoning via thought chains for science question answering," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[18] T. M. Haladyna, *Developing and Validating Multiple-Choice Test Items*, 3rd ed. Mahwah, NJ: Lawrence Erlbaum Associates, 2004.

[19] D. Coniam, "A preliminary inquiry into using corpus word frequency data in the automatic generation of English language cloze tests," *CALICO Journal*, vol. 14, no. 2–4, pp. 15–33, 1997.

[20] R. Mitkov and L. A. Ha, "Computer-aided generation of multiple-choice tests," in *Proceedings of the HLT-NAACL 03 Workshop on Building Educational Applications Using Natural Language Processing*, 2003, pp. 17–22.

[21] A. Papasalouros, K. Kanaris, and K. Kotis, "Automatic generation of multiple choice questions from domain ontologies," in *Proceedings of the IADIS International Conference e-Learning*, 2008.

[22] J. Martin Kowal, K. Hurley Bryant, D. Segall, and T. Kantrowitz, "Harnessing generative AI for assessment item development: Comparing AI-generated and human-authored items," *International Journal of Selection and Assessment*, vol. 33, no. 3, p. e70021, 2025.

[23] V. Raina and M. Gales, "Answer uncertainty and unanswerability in multiple-choice machine reading comprehension," in *Findings of the Association for Computational Linguistics: ACL 2022*, 2022.

[24] Z. Bai, P. Wang, T. Xiao, T. He, Z. Han, Z. Zhang, and M. Z. Shou, "Hallucination of multimodal large language models: A survey," 2025.

[25] C.-H. Chiang and H.-y. Lee, "Can large language models be an alternative to human evaluations?" in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023, pp. 15 607–15 631.

[26] D. Pugh, A. De Champlain, M. Gierl, H. Lai, and C. Touchie, "Can automated item generation be used to develop high quality MCQs that assess application of knowledge?" *Research and Practice in Technology Enhanced Learning*, vol. 15, no. 1, p. 12, 2020.

[27] D. Chen, R. Chen, S. Zhang, Y. Liu, Y. Wang, H. Zhou, Q. Zhang, Y. Wan, P. Zhou, and L. Sun, "MLLM-as-a-judge: Assessing multimodal LLM-as-a-judge with vision-language benchmark," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 235, 2024, pp. 6562–6595.

[28] X. Hu, M. Gao, S. Hu, Y. Zhang, Y. Chen, T. Xu, and X. Wan, "Are LLM-based evaluators confusing NLG quality criteria?" in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

[29] S. Lee, S. Park, and J. Im, "MM-JudgeBias: A benchmark for evaluating compositional biases in MLLM-as-a-judge," 2026, arXiv preprint.

[30] S. Park, J. Choi, J. Kang, S. Lee, J. Shin, and H. Shim, "Mitigating perceptual judgment bias in multimodal LLM-as-a-judge via perceptual perturbation and reward modeling," 2026, arXiv preprint.

[31] Z. Wang, G. Pang, Z. Liu, W. Miao, J. Zheng, and X. Bai, "On the adversarial robustness of multimodal LLM judges," 2026, arXiv preprint.

[32] X. Xie, J. W. K. Ho, C. Murphy, G. Kaiser, B. Xu, and T. Y. Chen, "Testing and validating machine learning classifiers by metamorphic testing," *Journal of Systems and Software*, vol. 84, no. 4, pp. 544–558, 2011.

[33] M. Papadakis, M. Kintis, J. Zhang, Y. Jia, Y. Le Traon, and M. Harman, "Mutation testing advances: An analysis and survey," in *Advances in Computers*. Elsevier, 2019, vol. 112, pp. 275–378.

[34] Y. Tian, K. Pei, S. Jana, and B. Ray, "DeepTest: Automated testing of deep-neural-network-driven autonomous cars," in *Proceedings of the 40th International Conference on Software Engineering (ICSE)*, 2018, pp. 303–314.

[35] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *International Conference on Learning Representations (ICLR)*, 2019.

[36] B. Yu, Z. Zhong, X. Qin, J. Yao, Y. Wang, and P. He, "Automated testing of image captioning systems," in *Proceedings of the 31st ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, 2022.

[37] B. Yu, Z. Zhong, J. Li, Y. Yang, S. He, and P. He, "ROME: Testing image captioning systems via recursive object melting," in *Proceedings of the 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, 2023.

[38] S. Tizpaz-Niari, V. Monjezi, M. Wagner, S. Darian, K. Reed, and A. Trivedi, "Metamorphic testing and debugging of tax preparation software," in *Proceedings of the 45th IEEE/ACM International Conference on Software Engineering: Software Engineering in Society (ICSE-SEIS)*, 2023.

[39] Y. Brun and A. Meliou, "Software fairness," in *Proceedings of the 26th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), New Ideas and Emerging Results*, 2018, pp. 754–759.

[40] S. Galhotra, Y. Brun, and A. Meliou, "Fairness testing: Testing software for discrimination," in *Proceedings of the 11th Joint Meeting on Foundations of Software Engineering (ESEC/FSE)*, 2017, pp. 498–510.

[41] Z. Chen, J. M. Zhang, M. Hort, M. Harman, and F. Sarro, "Fairness testing: A comprehensive survey and analysis of trends," *ACM Transactions on Software Engineering and Methodology*, vol. 33, no. 5, 2024.

[42] M. Bano, H. Gunatilake, and R. Hoda, "What does a software engineer look like? Exploring societal stereotypes in LLMs," in *Proceedings of the 47th IEEE/ACM International Conference on Software Engineering: Software Engineering in Society (ICSE-SEIS)*, 2025.

[43] A. Parziale, G. Voria, V. Pontillo, A. Di Salle, P. Pelliccione, G. Catolino, and F. Palomba, "Once upon a team: Investigating bias in LLM-driven software team composition and task allocation," in *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering: Software Engineering in Society (ICSE-SEIS)*, 2026.

[44] A. S. Luccioni, C. Akiki, M. Mitchell, and Y. Jernite, "Stable bias: Evaluating societal representations in diffusion models," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.

[45] S. Morales, R. Clarisó, and J. Cabot, "Automating bias testing of LLMs," in *Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2023, pp. 1705–1707.

[46] ——, "ImageBiTe: A framework for evaluating representational harms in text-to-image models," in *Proceedings of the IEEE/ACM 4th International Conference on AI Engineering – Software Engineering for AI (CAIN)*, 2025, pp. 95–106.

[47] P. Howard, A. Madasu, T. Le, G. Lujan Moreno, A. Bhiwandiwalla, and V. Lal, "SocialCounterfactuals: Probing and mitigating intersectional social biases in vision-language models with counterfactual examples," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[48] Holland *et al.*, "A qualitative exploration of student cognition when answering text-only or image-based histology multiple-choice questions," *Medical Science Educator*, 2024.

[49] R. E. Mayer, *Multimedia Learning*, 2nd ed. New York: Cambridge University Press, 2009.

[50] M. Hegarty, "The cognitive science of visual-spatial displays: Implications for design," *Topics in Cognitive Science*, vol. 3, no. 3, pp. 446–474, 2011.

[51] V. Crisp and E. Sweiry, "Can a picture ruin a thousand words? Physical aspects of the way exam questions are laid out," in *Annual Conference of the British Educational Research Association*, 2003.

[52] R. E. Bennett, "The changing nature of educational assessment," *Review of Research in Education*, vol. 39, no. 1, pp. 370–407, 2015.

[53] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.

[54] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York: Chapman & Hall, 1993.